

BUSINESS INTELLIGENCE PROJECT REPORT

AtliQ Hardware Sales Analytics Dashboard

An End-to-End SQL & Power BI Analytics Solution

Project Type:	Business Intelligence / Sales Analytics
Domain:	Sales & Business Analytics
Tools Used:	SQL • Power BI • Power Query • DAX
Primary Objective:	Analyze sales performance across markets, customers, products and time using an interactive Power BI dashboard.

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1. Project Overview

The **AtliQ Hardware Sales Analytics Dashboard** is an end-to-end Business Intelligence project developed using **SQL and Microsoft Power BI** to analyze sales performance across different markets, customers, products, and time periods.

The objective of the project was to transform raw sales transaction data into an interactive dashboard that enables business stakeholders to monitor revenue, sales quantity, market performance, customer contribution, and overall business trends.

The project covers the complete data analytics workflow, including **SQL data extraction, ETL, data transformation using Power Query, data modeling using a Star Schema, DAX-based calculations, interactive dashboard development, report navigation using bookmarks, and publishing the final report to Power BI Service.**

2. Business Problem

This learning project is based on the sales data and business scenario of **AtliQ Hardware**, a company operating across multiple markets and selling a range of hardware products to different customers.

The project was designed to explore how Business Intelligence tools can be used to analyze sales data and address common business reporting challenges, such as:

- Monitoring overall sales performance
- Comparing revenue across different markets
- Identifying high-performing and low-performing markets
- Analyzing sales quantity by market
- Understanding customer-level sales performance
- Tracking product-level performance
- Analyzing sales trends over time
- Identifying areas that may require management attention

To address these analytical requirements, an interactive Business Intelligence dashboard was developed using the available sales data. The project focuses on consolidating information from different dimensions into a single report and demonstrating how **SQL and Power BI** can be used to transform raw sales data into meaningful insights for business analysis and decision-making.

3. Project Objectives

The major objectives of the project were:

1. Extract and analyze sales data using SQL.
2. Clean and transform raw data using Power Query.

3. Implement an ETL workflow for preparing the data.
4. Build an efficient relational data model using a Star Schema.
5. Create calculated measures using DAX.
6. Develop interactive Power BI reports and dashboards.
7. Analyze revenue and sales quantity across different markets.
8. Enable users to filter and explore the data interactively.
9. Implement report navigation and interactive views using bookmarks.
10. Publish the completed dashboard to Power BI Service.

4. Tools & Technologies Used

Tool / Technology	Purpose
SQL Server Management Studio (SSMS) 22	Data extraction, querying and analysis
Power BI Desktop	Dashboard and report development
Power Query	Data cleaning, transformation and ETL
DAX	Measures, KPIs and analytical calculations
Star Schema	Data modeling
Power BI Service	Publishing and sharing the report
Bookmarks	Interactive navigation and report views

The project demonstrates the typical Power BI workflow of preparing data with Power Query, creating a dimensional model, developing DAX measures, and building business-focused reports. Microsoft recommends a fact-and-dimension structure such as a Star Schema for Power BI semantic models because it separates transactional data from descriptive dimensions and supports effective filtering and summarization.

5. Data Extraction Using SQL

SQL was used to interact with the underlying sales database and retrieve the required information for analysis.

The database contained information related to areas such as:

- Customers
- Products
- Markets

- Transactions
- Sales quantities
- Sales amounts
- Currency information

SQL queries were used to:

- Retrieve relevant datasets
- Join multiple tables
- Filter records
- Aggregate sales information
- Analyze market performance
- Validate data
- Investigate data quality issues

SQL therefore acted as an important part of the initial data extraction and analysis process.

6. ETL Process

The project followed an ETL approach:

Extract

Data was extracted from the SQL database.

Transform

The extracted data was processed using **Power Query**.

Transformation activities included:

- Removing unnecessary columns
- Handling missing or blank values
- Correcting data types
- Standardizing fields
- Transforming currency-related information
- Creating required columns
- Filtering irrelevant records
- Preparing tables for data modeling

Load

The transformed data was loaded into Power BI for modeling and analysis.

This workflow helped convert raw transactional data into a structured dataset suitable for business reporting.

7. Data Transformation Using Power Query

Power Query was used as the primary data preparation and transformation layer.

Key activities included:

- Data cleaning
- Column selection
- Data type conversion
- Filtering
- Handling null/blank values
- Creating conditional columns
- Data standardization
- Merging and shaping datasets
- Preparing dimension and fact tables

Power Query is particularly important in Power BI because the semantic model depends on data that has been appropriately prepared and transformed before analysis.

8. Data Modeling

A **Star Schema** was implemented to create a structured and efficient Power BI data model.

The model separates:

Fact Table

The fact table contains transactional sales information such as:

- Sales amount
- Sales quantity
- Product key
- Customer key
- Market key
- Date

Dimension Tables

Dimension tables provide descriptive information used to filter and analyze the sales transactions.

Examples include:

- Customer Dimension
- Product Dimension
- Market Dimension

- Date Dimension

The relationships between fact and dimension tables allow users to analyze sales from multiple business perspectives.

The Star Schema approach is widely recommended for Power BI because fact tables store business events/measures while dimension tables provide the attributes used for filtering and grouping.

9. DAX Measures

DAX (Data Analysis Expressions) was used to create analytical measures and KPIs.

Examples of measures developed for the dashboard include:

- Total Revenue
- Total Sales Quantity
- Revenue by Market
- Sales Quantity by Market
- Revenue by Customer
- Revenue by Product
- Revenue by Year
- Revenue by Month
- Profit-related metrics where applicable

DAX measures allow calculations to respond dynamically to slicers, filters, and report interactions.

For example:

```
Total Revenue = SUM(Sales[Sales Amount])
```

and:

```
Total Quantity = SUM(Sales[Sales Quantity])
```

These measures can then be analyzed dynamically by market, customer, product, or time period.

10. Dashboard Development

The Power BI report was designed to provide an interactive view of the company's sales performance.

The dashboard includes analytical views such as:

Revenue Analysis

Revenue performance can be analyzed across:

- Markets
- Customers

- Products
- Time periods

Sales Quantity Analysis

The report provides visibility into the quantity of products sold across different markets and other business dimensions.

Market Performance

Markets can be compared based on:

- Revenue
- Sales quantity
- Contribution to overall sales
- Historical performance

Trend Analysis

Time-based visuals help identify:

- Monthly sales trends
- Yearly performance
- Growth or decline patterns
- Seasonal variations

11. Interactive Features

Several Power BI features were implemented to improve report usability.

Slicers

Users can filter the report based on different dimensions such as:

- Market
- Customer
- Product
- Date
- Year
- Month

Bookmarks

Power BI Bookmarks were used to create interactive report states and improve navigation between different views.

This allows users to switch between different analytical perspectives without creating multiple independent reports.

Report Navigation

Navigation elements were implemented to make it easier for users to move between report pages and analytical views.

Interactive Visuals

Charts, tables, cards and other Power BI visuals were used to present the information in an easily understandable format.

12. Key Business Questions Addressed

The dashboard helps answer questions such as:

- What is the total revenue generated?
- How much quantity has been sold?
- Which markets generate the highest revenue?
- Which markets contribute the highest sales quantity?
- Which customers contribute most to revenue?
- Which products perform best?
- How does revenue change over time?
- Which markets are underperforming?
- What are the major sales trends?

13. Power BI Service & Publishing

After completing the dashboard in Power BI Desktop, the report was **published to Power BI Service**.

Publishing the report enabled the dashboard to be accessed through the Power BI online environment and demonstrated the complete workflow from:

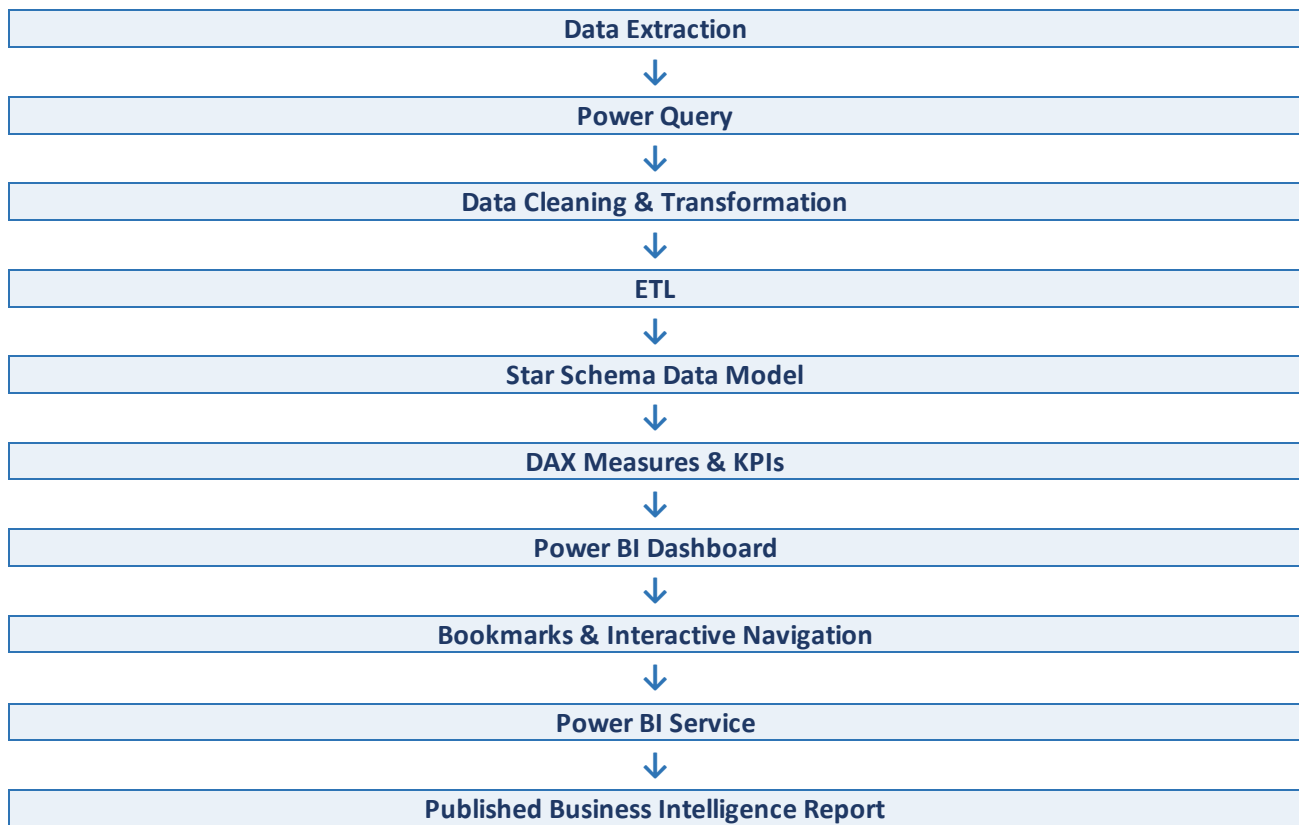
Data → Transformation → Modeling → Analysis → Visualization → Publishing

This represents an end-to-end BI reporting process rather than simply creating a static dashboard.

14. Project Workflow

The complete project workflow can be summarized as:





15. Key Skills Demonstrated

This project provided hands-on experience in:

Technical Skills

- SQL
- Power BI
- Power Query
- DAX
- ETL
- Data Cleaning
- Data Transformation
- Data Modeling
- Star Schema
- Data Visualization
- Power BI Service
- Report Publishing
- Bookmarks
- Interactive Dashboard Development

Analytical Skills

- Sales Performance Analysis
- Market Analysis
- Customer Analysis
- Product Analysis
- Trend Analysis
- KPI Development
- Business Reporting
- Data-driven Decision Making

16. Project Outcome

The final dashboard transformed raw sales data into an interactive Business Intelligence solution.

It provides management with a consolidated view of sales performance and enables users to quickly analyze **revenue and sales quantity across markets, customers, products and time periods**.

The project demonstrates the ability to take a dataset from the **data extraction stage through ETL, modeling, analysis, visualization and deployment**, which closely reflects the workflow used in real-world Business Intelligence projects.

17. Conclusion

The AtliQ Hardware Sales Analytics Dashboard demonstrates an end-to-end approach to Business Intelligence using SQL and Power BI.

The project combines **SQL querying, Power Query transformation, ETL, Star Schema modeling, DAX calculations, interactive visualization, bookmarks and Power BI Service publishing** to create a business-focused analytical solution.

More importantly, the project demonstrates that Power BI can be used not only to create attractive visualizations but also to transform raw business data into meaningful insights that support faster and more informed decision-making.

Technology Stack	SQL Power BI Power Query DAX ETL Star Schema Data Modeling Data Visualization Power BI Service Bookmarks
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